English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

English transcription of the handwritten solutions. Numbering and calculations follow the handwritten original.

1. Dividing couples into teams

1. Let X_1 count people in team A. Then

$$X_1 \sim \mathrm{Bin}(20, \tfrac{1}{2}), \qquad \boxed{\mathbb{E}X_1 = 10}.$$

Let X_2 count couples whose members are in the same team. Then

$$X_2 \sim \mathrm{Bin}(10, \tfrac{1}{2}), \qquad \boxed{\mathbb{E}X_2 = 5}.$$

2. A particular person is in the same team as their partner with probability $1/2$. The expected number of divisions is $\boxed{\mathbb{E}\,\mathrm{Ge}(\frac{1}{2}) = 2}$.

3. All couples stay together with probability $(1/2)^{10}$, so

$$\boxed{\mathbb{E}\,\mathrm{Ge}((\tfrac{1}{2})^{10}) = 1024}.$$

If X_A, X_B count couples in teams A and B, then $X_A \overset{d}{=} X_B$ and $X_A + X_B = 10$, hence $\boxed{\mathbb{E}X_A = 5}$.

2. Dice in the bag

Let H_i mean that we draw i fair and $2 - i$ biased dice; let A mean two sixes in two rolls. Initially,

$$\mathbb{P}(H_2) = \tfrac{1}{4}, \quad \mathbb{P}(H_1) = \tfrac{1}{2}, \quad \mathbb{P}(H_0) = \tfrac{1}{4}.$$

The likelihoods are

$$\mathbb{P}(A|H_2) = (\tfrac{1}{6})^2, \quad \mathbb{P}(A|H_1) = 0.4 \cdot \tfrac{1}{6}, \quad \mathbb{P}(A|H_0) = (0.4)^2.$$

Thus

$$\mathbb{P}(H_2|A) = \frac{(\frac{1}{6})^2 \frac{1}{4}}{(\frac{1}{6})^2 \frac{1}{4} + \frac{1}{2} \cdot 0.4 \cdot \frac{1}{6} + \frac{1}{4}(0.4)^2} = \frac{25}{289} = (\tfrac{5}{17})^2.$$

The handwritten solution leaves $\mathbb{P}(H_1|A)$ unevaluated and gives

$$\mathbb{P}(H_0|A) = \cdots = (\tfrac{12}{17})^2, \qquad \mathbb{P}(\text{biased die}|A) = \frac{12}{17} \approx 70.59\%.$$

We can also proceed in two stages. Let A now mean drawing a die and rolling six, H_0 drawing a fair die, and H_1 a biased one:

$$\mathbb{P}(H_0) = \mathbb{P}(H_1) = \tfrac{1}{2}, \quad \mathbb{P}(A|H_0) = \tfrac{1}{6}, \quad \mathbb{P}(A|H_1) = 40\%.$$

Then

$$\mathbb{P}(H_0|A) = \frac{\frac{1}{6}\frac{1}{2}}{\frac{1}{6}\frac{1}{2} + \frac{2}{5}\frac{1}{2}} = \frac{5}{17} \approx 29.41\%, \qquad \mathbb{P}(H_1|A) = \frac{12}{17}.$$

Write these updated hypotheses as H_0^*, H_1^*. Then

$$\mathbb{P}(H_0^*|A) = \frac{\frac{1}{6}\frac{5}{17}}{\frac{1}{6}\frac{5}{17} + \frac{2}{5}\frac{12}{17}} = \frac{25}{169} \approx 14.79\%,$$

and approximately 85.21% for a biased die.

Generalization. If we draw n times and roll only sixes, let H_i mean that i fair dice were drawn:

$$\mathbb{P}(H_i) = \binom{n}{i}(\tfrac{1}{2})^n, \qquad \mathbb{P}(A|H_i) = (\tfrac{1}{6})^i(\tfrac{2}{5})^{n-i}.$$

Hence

$$\mathbb{P}(H_n|A) = \frac{(\frac{1}{6})^n(\frac{1}{2})^n}{\sum_{i=0}^{n}\binom{n}{i}(\frac{1}{2})^n(\frac{1}{6})^i(\frac{2}{5})^{n-i}} = \frac{(\frac{1}{6})^n}{(\frac{17}{30})^n} = (\tfrac{5}{17})^n.$$

The notes ask: $\mathbb{P}(\text{fair die}) = 5/17$?! This is a procedure for updating beliefs after a new draw.

3. Penalty shootout

1. In each of the first five rounds, both teams must score or both miss:

$$(75\% \cdot 80\% + 25\% \cdot 20\%)^5 \approx 11.6\%.$$

2. Goals in the first five rounds are $X \sim \mathrm{Bin}(5, 80\%)$ for team 1 and $Y \sim \mathrm{Bin}(5, 75\%)$ for team 2. Thus

$$\mathbb{P}(X = Y) = \sum_{k=0}^{5} \mathbb{P}(X = Y = k) = \sum_{k=0}^{5} \binom{5}{k}^2 (75\% \cdot 80\%)^k (25\% \cdot 20\%)^{5-k} = 0.2984.$$

   The probability of a tie after ten rounds is

$$\mathbb{P}(X = Y)(75\% \cdot 80\% + 25\% \cdot 20\%)^5 \approx 3.46\%.$$

3. The handwritten expected-penalty calculation is

$$10(1 - \mathbb{P}(X = Y)) + \mathbb{P}(X = Y)\left(10 + \frac{1}{1 - (75\% \cdot 80\% + 25\% \cdot 20\%)}\right) \approx 10.85.$$

4. Breaking the segment

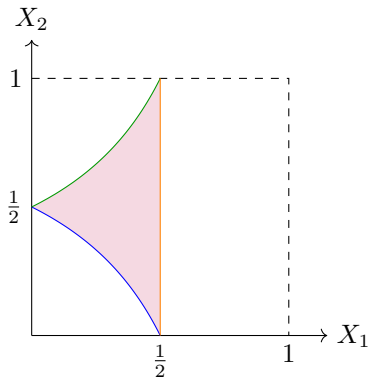
Choose $X_1 \sim U(0,1)$ and, conditionally, $Y \sim U(0, 1 - X_1)$. Write $Y = (1 - X_1)X_2$ with independent $X_1, X_2 \sim U(0,1)$. The three lengths are $X_1, Y, 1 - X_1 - Y$. The triangle inequalities become

$$\begin{cases} X_1 + Y > 1 - X_1 - Y, \\ X_1 + (1 - X_1 - Y) > Y, \\ Y + (1 - X_1 - Y) > X_1 \end{cases} \iff \begin{cases} X_1 + Y > \frac{1}{2}, \\ Y < \frac{1}{2}, \\ X_1 < \frac{1}{2}. \end{cases}$$

Substituting $Y = (1 - X_1)X_2$ gives

$$\begin{cases} X_1 + (1 - X_1)X_2 > \frac{1}{2}, \\ (1 - X_1)X_2 < \frac{1}{2}, \\ X_1 < \frac{1}{2} \end{cases} \iff \begin{cases} X_2 > \dfrac{1 - 2X_1}{2(1 - X_1)}, \\ X_2 < \dfrac{1}{2(1 - X_1)}, \\ X_1 < \frac{1}{2}. \end{cases}$$

The favourable region lies between these curves and left of $X_1 = 1/2$:



The required area ratio is

$$\frac{S_{\text{favourable}}}{S_\Omega} = \frac{\displaystyle\int_0^{1/2} \left(\frac{1}{2(1-x)} - \frac{1-2x}{2(1-x)}\right) dx}{1} = \int_0^{1/2} \frac{x}{1-x}\, dx = \ln 2 - \frac{1}{2} \approx 19.32\%.$$